BRUNVOLL

Technical Report

Voyage Fuel Optimiser – Software Architecture & Validation

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1 Introduction

This report documents the architecture and validation of the `voyage_comparison` propulsion optimiser tool. The tool performs long-term hindcast-based voyage simulations comparing the factory combinator pitch/RPM schedule against a fuel-optimal operating point selection, with optional Flettner rotor wind-assist.

The tool was originally implemented as a single 4 913-line Python script. A modular refactoring was carried out to improve maintainability, testability, and extensibility. This report describes the physical models, the software architecture after refactoring, and presents validation results from a full-year simulation on the Rotterdam–Gothenburg route.

1.1 Scope

The tool addresses the following question: *for a given vessel on a fixed route, how much fuel can be saved by optimising the propeller pitch and RPM at each operating point, compared to the factory-default combinator schedule?*

The analysis chain covers:

- Calm-water hull resistance from model test data.
- Wave added resistance via PdStrip drift transfer functions integrated over JONSWAP spectra.
- Blendermann (1994) wind resistance model.
- Flettner rotor wind-assist thrust (Bordogna et al., 2019).
- Hull and propeller roughness penalties (Townsin–Dey, 2003).
- Factory combinator schedule replication.
- Fuel-optimal pitch/RPM selection via propeller model optimisation.

Hindcast weather is provided by NORA3 (Norwegian Reanalysis Archive, 3 km resolution) wave and wind fields.

2 Physical Models

2.1 Hull Resistance

Calm-water resistance is interpolated from the vessel 206 model test trial prediction (test 25-0461/25-0288, draught $7.600\,\mathrm{m}$, displacement $11\,165\,\mathrm{m}^3$) at 15 speeds ($8\,\mathrm{kn}$ to $15.0\,\mathrm{kn}$). No sea margin removal is needed — the resistance values are clean calm-water. The total thrust demand including environmental loads is:

$$T_{\text{prop}} = \frac{R_{\text{calm}} + R_{\text{aw}} + R_{\text{wind}} + R_{\text{rough}} - F_{\text{Flettner}}}{1 - t} \tag{1}$$

Wake fraction $w(V_s)$ and thrust deduction $t(V_s)$ are interpolated from the self-propulsion test data.

2.2 Wave Added Resistance

Wave added resistance is computed from PdStrip strip-theory transfer functions. The `DriftTransferFunction` class loads the `.dat` output from PdStrip and computes the mean surge drift force for a given sea state:

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$$R_{\text{aw}} = 2\int_0^\infty S_\zeta(\omega;\, H_s, T_p)\cdot C_{\text{drift}}(\omega,\, \theta_{\text{rel}})\, d\omega \tag{2}$$

where S_ζ is the JONSWAP wave spectrum and C_{drift} is the frequency- and heading-dependent drift force transfer function per unit wave amplitude squared.

2.3 Wind Resistance

Hull wind resistance follows Blendermann (1994), using tabulated surge force coefficients $C_X(\alpha)$ for a general cargo vessel:

$$R_{\text{wind}} = -C_X(\alpha_{\text{app}})\cdot \tfrac{1}{2}\,\rho_{\text{air}}\,V_{\text{app}}^2\cdot A_F \tag{3}$$

where α_{app} is the apparent wind angle from the bow, V_{app} the apparent wind speed, and $A_F = 280\,\mathrm{m}^2$ the frontal projected wind area.

2.4 Flettner Rotor

The Flettner rotor model is ported from the BruCon C++ vessel simulator. Lift and drag coefficients follow Bordogna et al. (2019) with aspect-ratio correction for a rotor of height $28\,\mathrm{m}$ and diameter $4\,\mathrm{m}$:

- Lift and drag are tabulated as functions of spin ratio SR $= \Omega R/V_{\text{app}}$.
- The target spin ratio is SR $= 3$ (max $220\,\mathrm{RPM}$).
- Rotor drive power follows the Theodorsen & Regier (1944) model.
- Rotor motor fuel is charged at the auxiliary genset SFOC ($215\,\mathrm{g}\cdot\mathrm{h/kW}$).

The net propulsive benefit is the surge component of the Magnus-effect force minus the fuel penalty for driving the rotor.

2.5 Hull and Propeller Roughness

Hull roughness resistance follows Townsin & Dey (2003):

$$\Delta C_F = 0.044\left[\left(\frac{k_s}{L}\right)^{1/3} - 10\,\text{Re}^{-1/3}\right] + 0.000125 \tag{4}$$

Propeller blade roughness increases torque (and thus fuel) via the empirical model of Mosaad (1986). Both roughness values can be specified directly in μm or via a `FoulingScenario` which models piecewise-linear roughness growth over time since last cleaning.

2.6 Factory Combinator

The `FactoryCombinator` class replicates the BruCon vessel simulator's `MakeNormalCombinator()` algorithm. For a given thrust demand, it returns the pitch and RPM dictated by the factory schedule—following the propeller curve up to a break point, then constant RPM with increasing pitch.

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2.7 Propeller Model and Fuel Optimisation

The fuel-optimal operating point is found by searching the full $(P/D,\, n)$ space of the C-series propeller model. For each thrust demand, the optimiser finds the pitch/RPM combination that minimises engine SFOC $\times$ brake power, subject to the MAN L27/38 power envelope.

Results are pre-computed for a grid of 681 thrust values ($10\,\mathrm{kN}$ to $350\,\mathrm{kN}$ in $0.5\,\mathrm{kN}$ steps) using parallel workers, then looked up during the voyage simulation for speed.

3 Software Architecture

3.1 Overview

The original implementation was a single file (`voyage_comparison.py`, 4 913 lines). The refactoring decomposed it into four sub-packages with a thin CLI entry point:

```
voyage_comparison.py       323 lines   CLI entry point (argparse)
models/                  1,500 lines   Physical models & data
simulation/              1,117 lines   Simulation engine & orchestration
plotting/                1,218 lines   Matplotlib visualisations
reporting/               1,023 lines   Text-based summary reports
                         -----
Total                    5,181 lines
```

3.2 `models/` Package

Contains all physical models, vessel data, and environmental data access.

Module	Lines	Description
`constants.py`	129	File paths, propeller/engine parameters, hull data arrays, Blendermann coefficient tables
`roughness.py`	268	`FoulingScenario` dataclass, Townsin–Dey hull roughness, blade roughness factor
`combinator.py`	256	`FactoryCombinator` – factory pitch/RPM schedule
`drift_force.py`	203	`DriftTransferFunction` – PdStrip wave added resistance
`flettner.py`	176	`FlettnerRotor` – Magnus-effect wind-assist model
`route.py`	168	`Waypoint`, `Route` – great-circle navigation
`weather.py`	150	`WeatherAlongRoute` – NORA3 NetCDF data loading
`wind_resistance.py`	108	Blendermann (1994) hull wind resistance
`geometry.py`	17	`angle_diff()` helper shared across models
`__init__.py`	35	Re-exports for convenient importing

3.3 `simulation/` Package

Contains the core simulation loop, caching infrastructure, and top-level orchestration functions.

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Module	Lines	Description
`engine.py`	445	`evaluate_voyage()` – hourly stepping loop, round-trip combiner, parallel worker
`orchestrator.py`	425	`run_annual_comparison()`, `run_speed_sweep()`, `run_scheduling_analysis()`
`cache.py`	131	Pre-computed thrust-to-fuel lookup tables (parallel build)
`results.py`	108	`HourlyResult`, `VoyageResult`, `SpeedSweepResult` dataclasses
`__init__.py`	13	Re-exports

The core simulation loop in `evaluate_voyage()` proceeds as follows:

1. Interpolate the route at 1-hour intervals (great-circle).
2. Compute speed-dependent quantities once: wake fraction, thrust deduction, calm-water resistance, hull roughness resistance, blade roughness factor.
3. For each hourly position:
   - Fetch NORA3 weather (H_s, T_p, wave/wind direction and speed).
   - Compute wave added resistance via drift transfer function.
   - Compute Flettner rotor thrust and power.
   - Compute Blendermann wind resistance.
   - Assemble total thrust demand: $T = (R_{\text{calm}} + R_{\text{aw}} + R_{\text{wind}} + R_{\text{rough}} - F_{\text{Flettner}})/(1-t)$.
   - Evaluate factory combinator and optimiser fuel rates (from cache).
   - Also evaluate no-Flettner variants for savings decomposition.
4. Aggregate fuel totals, weather means, and compute savings split.

Only hours where all four evaluations (factory $\pm$ Flettner, optimiser $\pm$ Flettner) are feasible are counted, ensuring a consistent comparison basis.

3.4 `plotting/` Package

Module	Lines	Description
`comparison.py`	481	Annual comparison time-series and histogram plots
`departure.py`	192	Departure timing flexibility analysis plots
`scheduling.py`	191	2D speed $\times$ departure window heatmaps
`map_plot.py`	189	Route map with Natural Earth coastlines
`speed_sweep.py`	163	Speed sensitivity sweep plots
`__init__.py`	8	Re-exports

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3.5 `reporting/` Package

Module	Lines	Description
`summary.py`	519	`print_summary()`, `print_departure_analysis()`
`scheduling.py`	390	`print_scheduling_analysis()`, `print_roughness_sweep_summary()`
`speed_sweep.py`	106	`print_speed_sweep_summary()`
`__init__.py`	12	Re-exports

3.6 CLI Interface

The entry point `voyage_comparison.py` exposes 29 command-line arguments via `argparse`. Key modes of operation:

Flag	Description
`--download`	Pre-download NORA3 data for the route
`--year YEAR`	Hindcast year (default: 2024)
`--speed SPEED`	Transit speed in knots (default: 10)
`--plot`	Generate summary plots
`--no-flettner`	Disable Flettner rotor
`--compare-sg`	Standard vs shaft generator comparison
`--speed-sweep KN ...`	Speed sensitivity sweep
`--roughness-sweep T ...`	Fouling age sensitivity sweep
`--departure-analysis`	Departure timing flexibility analysis
`--scheduling-analysis KN ...`	2D departure $\times$ speed optimisation
`--hull-roughness KS_UM`	Hull roughness k_s [\$$\mu$\$m]
`--fouling-years T`	Time since cleaning [years]

4 Validation Results

A full-year simulation was run on the Rotterdam–Gothenburg route for 2024 as an end-to-end validation of the refactored codebase:

```
python voyage_comparison.py --year 2024 --quiet
```

4.1 Weather Data

The simulation uses wave and wind data from NORA3, the third-generation Norwegian Reanalysis Archive produced by MET Norway. A *hindcast* is a numerical weather simulation run retrospectively: the atmospheric and wave models are forced with observed boundary conditions and assimilate available measurements, producing a physically consistent, gap-free reconstruction of historical weather on a regular grid. Unlike point measurements from buoys or ships, a hindcast provides complete spatial and temporal coverage, making it suitable for route-based voyage simulation.

NORA3 uses the WAM spectral wave model driven by the HARMONIE-AROME atmospheric model. The wave data used here covers the North Sea and Skagerrak on a rotated-pole grid with approximately $3\,\mathrm{km}$ spatial resolution (Δlat $\approx 0.02°$, Δlon $\approx 0.035°$, corresponding to $\approx 2.2\,\mathrm{km} \times 2.2\,\mathrm{km}$ at the route latitude) and hourly temporal resolution. The data spans 1993 to the present.

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The variables extracted for each route point are:

- H_s – significant wave height [m]
- T_p – peak wave period [s]
- θ_q – mean wave direction [deg, meteorological convention]
- U_{10} – 10 m wind speed [m/s]
- θ_w – wind direction [deg, meteorological convention]

Monthly files are pre-downloaded for a bounding box around the route ($\pm 0.5^\circ$ margin) and stored as NetCDF files. During simulation, the weather at each hourly vessel position is obtained by nearest-neighbour lookup in both space and time. Given the $3\,\mathrm{km}$ grid spacing, the maximum spatial interpolation error is approximately $1.5\,\mathrm{km}$ – well within the spatial decorrelation scale of the wave field. The hourly time step matches the simulation stepping interval exactly, so temporal interpolation is not required.

4.2 Simulation Setup

Parameter	Value	Unit
Route	509	nm (one-way)
Transit speed	10	kn
Transit time (one-way)	50.9	hours
Mode	Round-trip	
Flettner rotor	Enabled	
Hull roughness	Clean (no fouling)	
Blade roughness	Clean (no fouling)	
Idle/port time	15	%
Fuel price	650	EUR/tonne
Propeller	$D = 4.80$	m
Engine	MAN L27/38	8-cyl, 800 RPM

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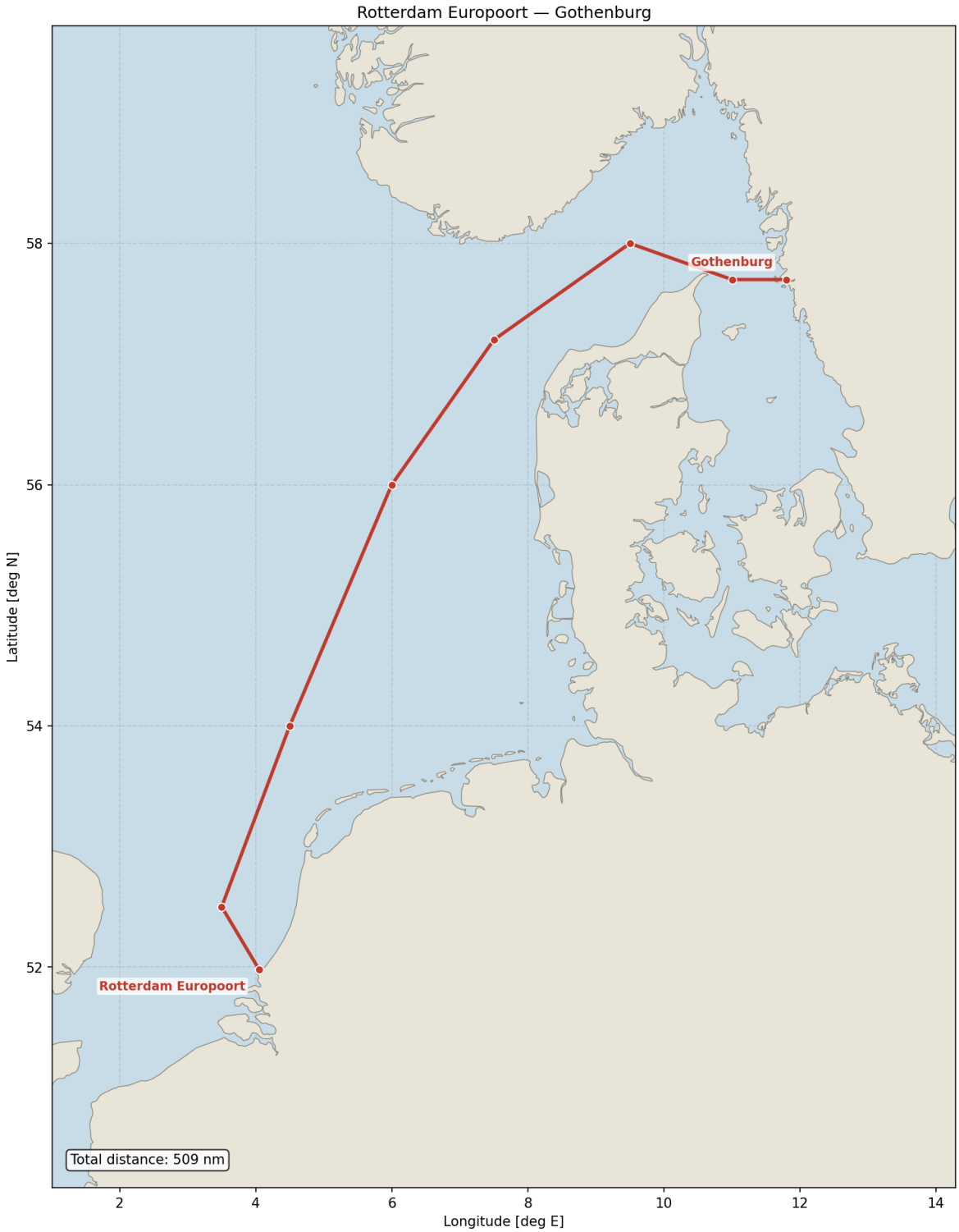

Figure 4-1: Rotterdam–Gothenburg route (509 nm one-way) with waypoints and leg bearings. Coastline data from Natural Earth 10m.

4.3 Voyage Statistics

362 round-trip voyages were successfully simulated (4 late-December departures failed due to missing January 2025 weather data).

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Metric	Mean	Median	P10	P90	Min	Max
Factory no-Fl [kg/voy]	17271	17062	16107	18996	12980	22239
Opt + Flettner [kg/voy]	15000	15043	12865	16959	10253	20844
Pitch/RPM saving [kg/voy]	899	929	794	970	462	998
Flettner saving [kg/voy]	1372	1208	331	2719	8	4942
Pitch/RPM saving	5.2%	5.4%	4.2%	5.9%	2.6%	6.4%
Flettner saving	8.0%	6.9%	2.0%	16.1%	0.0%	28.9%
Mean H_s [m]	1.4	1.3	0.7	2.3	0.4	4.0
Mean wind speed [m/s]	7.8	7.6	5.3	10.6	3.5	14.5
Mean R_{aw} [kN]	4.7	2.1	0.2	12.4	-0.9	45.2
Mean R_{wind} [kN]	4.5	4.0	-0.6	10.5	-4.5	26.9
Mean Flettner thrust [kN]	14.3	11.5	3.2	30.5	0.1	55.7
Mean rotor power [kW]	17.4	13.9	4.0	37.6	0.2	68.0

Fuel savings are decomposed additively against the factory-no-Flettner baseline:

$$\underbrace{\text{Factory}_{\text{NF}} - \text{Opt}_{+\text{Fl}}}_{\text{combined saving}} = \underbrace{\text{Factory}_{\text{NF}} - \text{Opt}_{\text{NF}}}_{\text{pitch/RPM}} + \underbrace{\text{Opt}_{\text{NF}} - \text{Opt}_{+\text{Fl}}}_{\text{Flettner}} \tag{5}$$

The pitch/RPM saving (5.2%) isolates the benefit of optimising the operating point at the same thrust demand. The Flettner saving (7.9%) isolates the thrust reduction from the rotor. Both percentages are relative to the factory-no-Flettner baseline.

Figure 4-2 shows the daily evolution of fuel savings, significant wave height, and force components across the full year.

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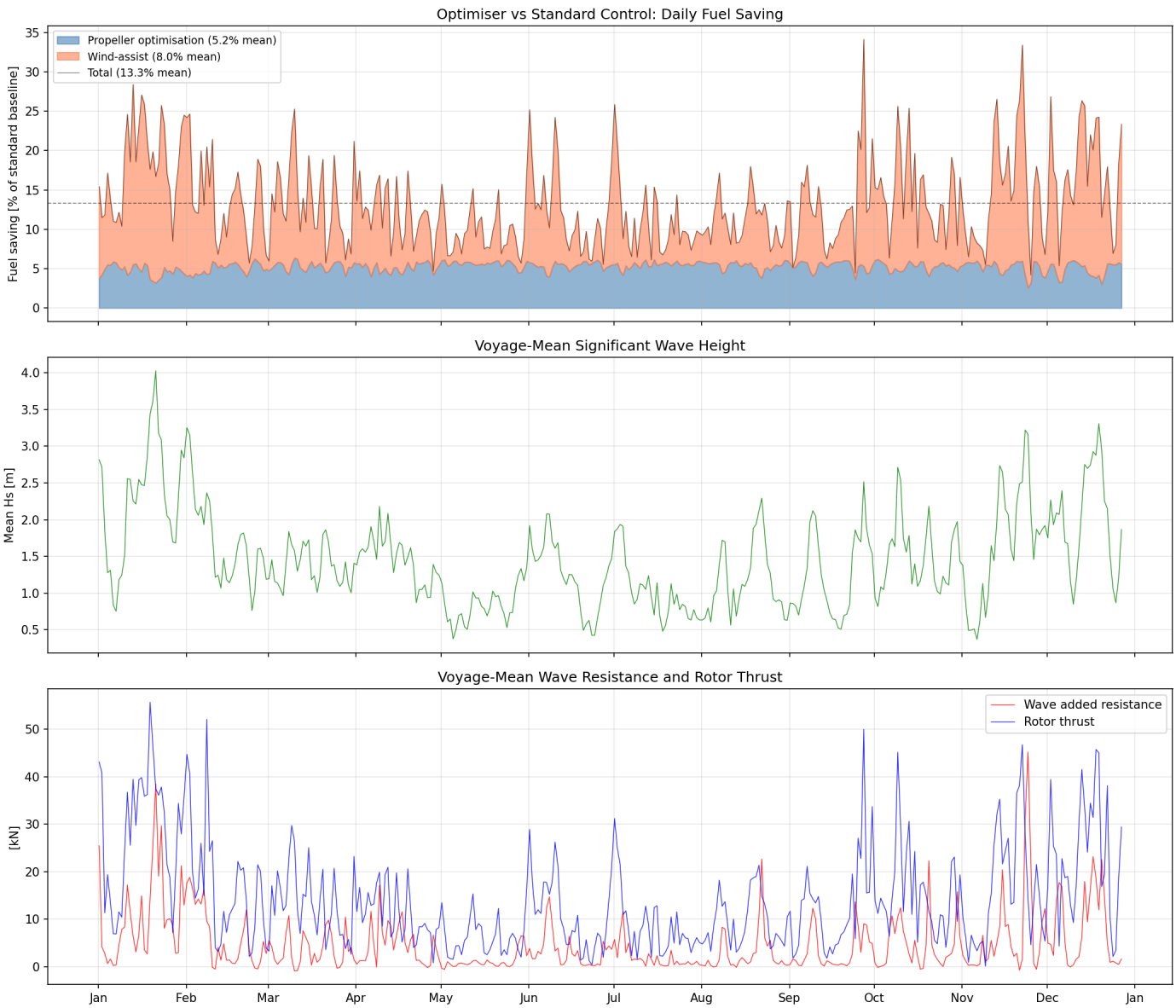

Figure 4-2: Annual time-series of per-voyage results. Top: pitch/RPM and Flettner fuel savings (stacked). Middle: significant wave height. Bottom: added wave resistance, wind resistance, and Flettner thrust.

4.4 Factory Combinator Feasibility

Metric	Value
Total evaluation hours	37 648
Both feasible	98%
Factory infeasible hours	0%
Voyages with ≥ 1 infeasible hour	7%

Factory infeasibility occurs when the engine exceeds its power envelope at the factory combinator's fixed RPM schedule. The optimiser adapts pitch/RPM to remain feasible.

Figure 4-3 shows the correlation between per-voyage fuel saving and mean sea state, coloured by mean wind speed. Higher seas drive larger absolute savings through the Flettner rotor.

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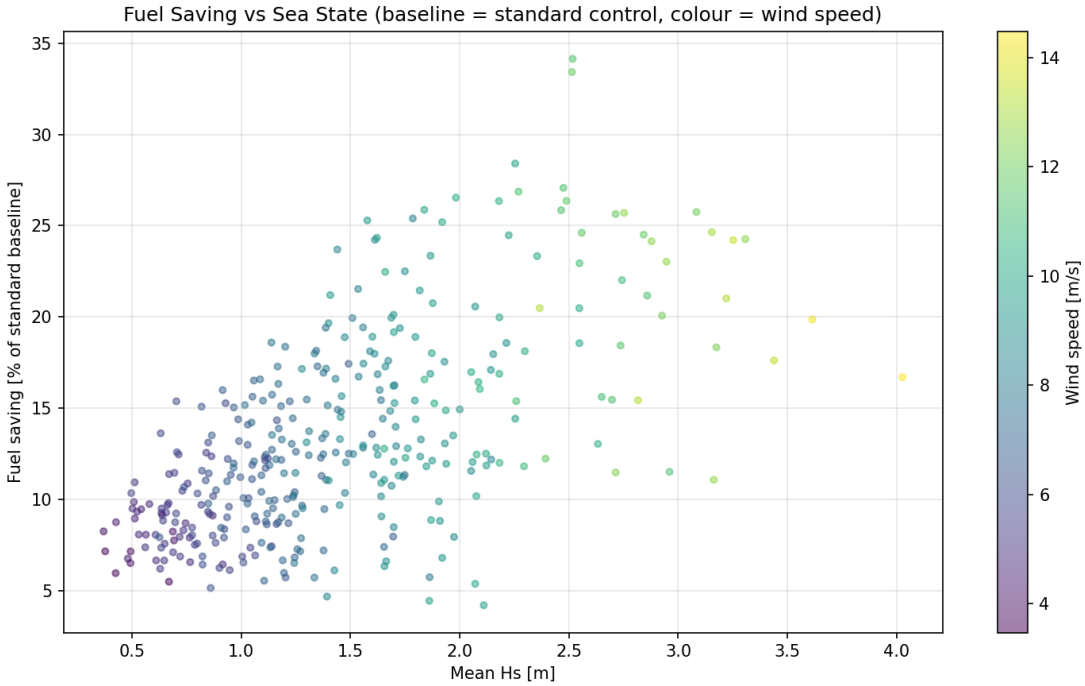

Figure 4-3: Per-voyage fuel saving vs mean significant wave height, coloured by mean wind speed. The positive correlation reflects the Flettner rotor's increasing benefit in windier conditions.

4.5 Annualised Fuel Consumption

Assuming 15% idle time, the vessel completes approximately 73 round-trip voyages per year.

Configuration	Fuel [tonnes/year]
Factory, no Flettner	1262.9
Factory + Flettner	1161.8
Optimiser, no Flettner	1197.2
Optimiser + Flettner	1096.9

Saving component	Tonnes/year	Percentage
Pitch/RPM	65.7	5.2%
Flettner	100.3	7.9%
Combined	166.1	13.1%

Figure 4-4 shows the distribution of per-voyage savings, decomposed into pitch/RPM optimisation and Flettner wind-assist components.

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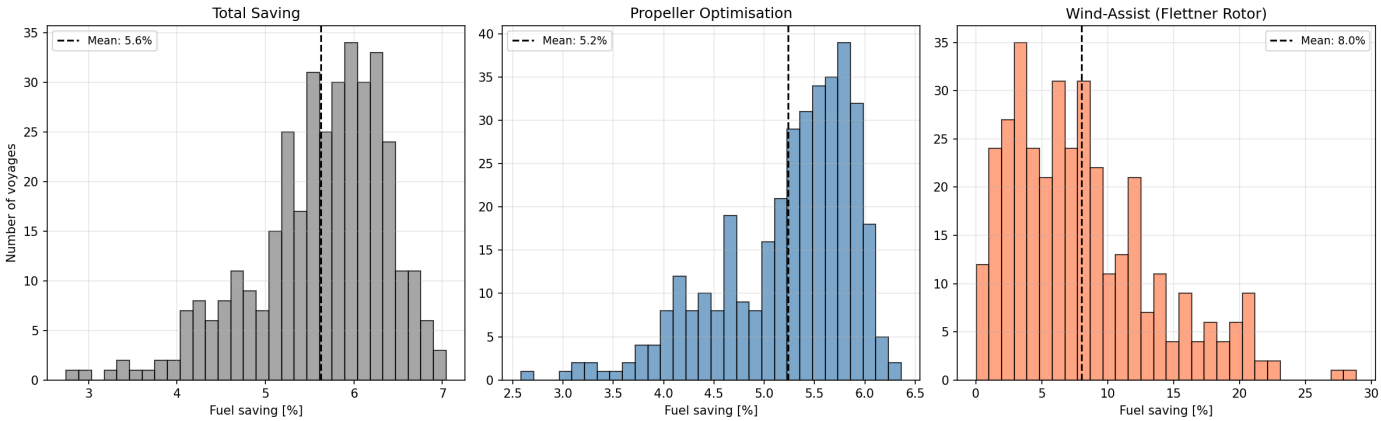

Figure 4-4: Histograms of per-voyage fuel savings. Left: total saving (factory vs optimiser+Flettner). Centre: pitch/RPM component. Right: Flettner component. The Flettner saving has a wider spread reflecting its weather dependence.

4.6 Cost Estimates

At MGO price of €650/tonne (typical ECA zone):

Configuration	Annual cost
Factory, no Flettner	€820 911
Factory + Flettner	€755 157
Optimiser, no Flettner	€778 194
Optimiser + Flettner	€712 970
Annual saving	€107 941 (13.1%)

4.7 Seasonal Breakdown

Season	Voyages	Fac NF [kg]	Opt+Fl [kg]	Saving	P/R	Flettner	Mean H_s [m]
Q1 (Jan–Mar)	91	17 327	14 731	5.4%	5.0%	10.1%	1.8
Q2 (Apr–Jun)	91	17 315	15 378	5.8%	5.4%	5.9%	1.2
Q3 (Jul–Sep)	92	17 181	15 206	5.8%	5.4%	6.1%	1.1
Q4 (Oct–Dec)	88	17 261	14 671	5.5%	5.1%	10.1%	1.7

The Flettner benefit is roughly doubled in the winter quarters (Q1, Q4) compared to summer, reflecting the stronger and more consistent North Sea wind regime. The pitch/RPM optimisation saving is relatively stable across seasons at 5.0–5.4%, as it depends primarily on the thrust demand distribution rather than its magnitude.

4.8 Shaft Generator Mode

The preceding results assume the vessel operates without a shaft generator (SG). When the SG is online, two additional constraints apply:

1. **Parasitic load:** The SG extracts electrical power from the main engine shaft, increasing total engine load. A typical average SG load of $150\,\mathrm{kW}$ is assumed.

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2. **RPM floor:** The SG must maintain output frequency within the permitted band ($48\,\mathrm{Hz}$ to $60\,\mathrm{Hz}$). At the PTO gear ratio, this constrains the minimum engine RPM to approximately $640\,\mathrm{RPM}$ (shaft RPM $\geq 94.1\,\mathrm{RPM}$).

Both constraints apply equally to the factory combinator and the optimiser. The factory combinator schedule is constructed with $200\,\mathrm{kW}$ SG headroom (the margin the hydrodynamicist uses when designing the combinator, on top of the standard $300\,\mathrm{kW}$ between the propeller curve and the torque limit), while the actual SG load of $150\,\mathrm{kW}$ is used in the fuel evaluation for both factory and optimiser.

```
python voyage_comparison.py --year 2024 --speed 10 --quiet \
    --sg-load 150 --sg-factory-allowance 200 \
    --sg-freq-min 48 --sg-freq-max 60
```

4.8.1 Annualised Fuel (SG Mode)

Configuration	Fuel [tonnes/year]
Factory, no Flettner	1513.9
Factory + Flettner	1414.9
Optimiser, no Flettner	1437.7
Optimiser + Flettner	1332.0

Saving component	Tonnes/year	Percentage
Pitch/RPM	76.3	5.0%
Flettner	105.7	7.0%
Combined	182.0	12.0%

4.8.2 Interpretation

With the corrected hull data (model test trial prediction), the calm-water thrust demand at $10\,\mathrm{kn}$ is approximately $133\,\mathrm{kN}$, requiring shaft power of approximately $767\,\mathrm{kW}$ and shaft RPM ≈ 75. Even with the SG RPM floor at $94.1\,\mathrm{RPM}$, the factory combinator schedule is built with the SG headroom, so its operating points differ from the optimiser's. The higher resistance means the engine operates well above the SG RPM floor for much of the voyage, giving the optimiser room to select a better pitch/RPM combination.

The pitch/RPM saving is 5.0%, comparable to the 5.2% in standard mode. The Flettner saving is 7.0%.

At $10\,\mathrm{kn}$ the SG constraint does not significantly affect savings with the corrected hull data. The speed sweep under SG constraints was not re-run with the updated hull data; however, at $10\,\mathrm{kn}$ the results are:

Speed	Pitch/RPM saving	Flettner saving	Combined saving	Annual saving
$10\,\mathrm{kn}$	5.0%	7.0%	12.0%	€118k

4.8.3 SG vs Auxiliary Genset Fuel Efficiency

An important question is whether the shaft generator is the most fuel-efficient way to supply $150\,\mathrm{kW}$ of electrical power. The alternative is a separate auxiliary diesel genset.

With the SG on the main engine, the factory combinator burns $1514\,\mathrm{t/yr}$. Without the SG, it burns $1263\,\mathrm{t/yr}$ — a difference of $251\,\mathrm{t/yr}$. This incremental fuel for $150\,\mathrm{kW}$ over $7439\,\mathrm{h/yr}$ corresponds to an *effective* SFOC of $225\,\mathrm{g/kWh}$.

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This is only slightly worse than a typical auxiliary genset (210 –-$230\,\mathrm{g/kWh}$). With the corrected hull data, the engine operates at higher load even at $10\,\mathrm{kn}$, reducing the RPM floor penalty. The effective SFOC is close to the genset range, making the SG vs genset choice marginal at this speed.

The breakeven genset SFOC — the value at which a separate genset becomes more expensive than running the SG on the ME — is $225\,\mathrm{g/kWh}$ for the factory combinator. At a genset SFOC of $215\,\mathrm{g/kWh}$, a separate auxiliary genset saves approximately $11\,\mathrm{t/yr}$ (€7k/yr) compared to the SG — a much smaller margin than previously estimated.

At higher speeds the picture changes: the RPM constraint penalty diminishes further, and the ME's superior part-load SFOC makes the SG more competitive.

4.8.4 Cost Estimates (SG Mode)

At MGO €650/tonne:

Configuration	Annual cost
Factory, no Flettner	€984 050
Optimiser + Flettner	€865 769
Annual saving	€118 281 (12.0%)

4.8.5 Seasonal Breakdown (SG Mode)

Season	Voyages	Fac NF [kg]	Opt+Fl [kg]	Saving	P/R	Flettner	Mean H_s [m]
Q1 (Jan–Mar)	91	20 713	17 886	5.9%	4.9%	8.9%	1.8
Q2 (Apr–Jun)	91	20 728	18 587	5.9%	5.2%	5.2%	1.2
Q3 (Jul–Sep)	92	20 581	18 400	5.9%	5.3%	5.4%	1.1
Q4 (Oct–Dec)	88	20 794	17 975	6.0%	4.9%	8.8%	1.7

4.8.6 Summary Comparison: Standard vs SG Mode

Metric	Standard mode	SG mode (150 kW)
Baseline fuel [t/yr]	1262.9	1513.9
Pitch/RPM saving	5.2% (66 t/yr)	5.0% (76 t/yr)
Flettner saving	7.9% (100 t/yr)	7.0% (106 t/yr)
Combined saving	13.1% (166 t/yr)	12.0% (182 t/yr)
Annual cost saving	€108k	€118k

The choice of whether to operate with the SG has a modest impact on savings. When the SG is offline, the Optimiser saves 5.2% from pitch/RPM optimisation. When the SG is online, the higher resistance from the corrected hull data means the engine operates well above the RPM floor, preserving most of the optimiser's freedom — the pitch/RPM saving is 5.0%. The Flettner rotor remains beneficial in both cases with similar relative impact.

The effective SFOC of the main engine for the SG load is $225\,\mathrm{g/kWh}$, close to a typical auxiliary genset ($215\,\mathrm{g/kWh}$). At $10\,\mathrm{kn}$ transit speed, a separate genset is only marginally more fuel-efficient than the shaft generator, saving approximately $11\,\mathrm{t/yr}$ (€7k/yr).

Figure 4-5 shows the side-by-side comparison of daily fuel savings in standard and SG modes.

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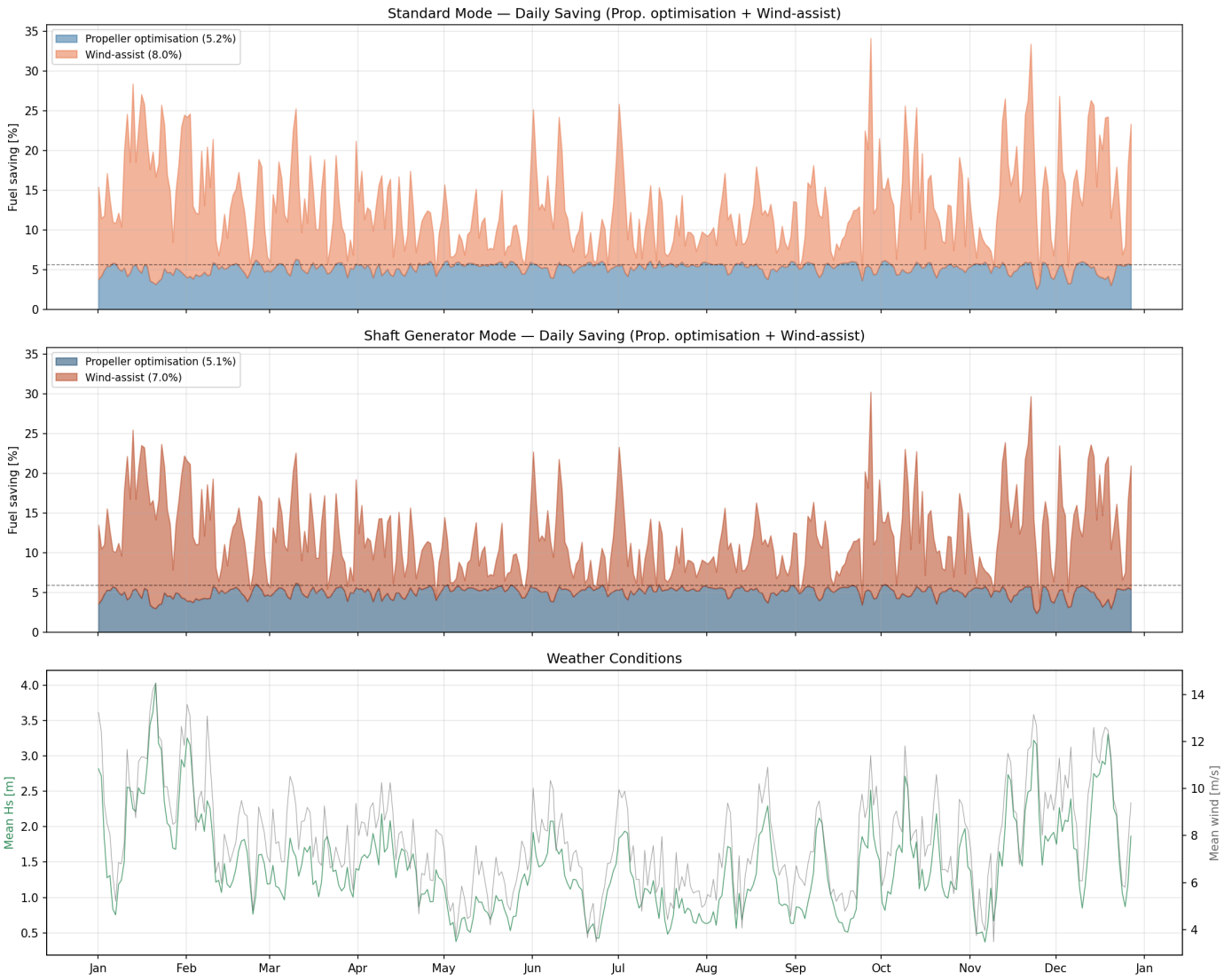

Figure 4-5: Side-by-side comparison of daily fuel savings in standard mode (top) vs shaft generator mode (bottom). With the corrected hull data, the pitch/RPM saving is preserved in SG mode, and the Flettner benefit remains comparable.

5 Conclusions

The refactored `voyage_comparison` tool has been validated by running a full-year (2024) hindcast simulation on the Rotterdam–Gothenburg route. Key findings:

1. **Standard mode (SG offline):** Pitch/RPM optimisation saves **5.2%** (66 t/yr, €43k/yr); Flettner wind-assist adds **7.9%** (100 t/yr, €65k/yr); combined **13.1%** (166 t/yr, €108k/yr) at $10\,\mathrm{kn}$.
2. **SG mode (150 kW, 48–60 Hz):** With the corrected hull data (higher resistance), the engine operates well above the SG RPM floor even at $10\,\mathrm{kn}$, preserving most of the optimiser's freedom. Pitch/RPM saving is **5.0%** (76 t/yr). Flettner retains **7.0%** (106 t/yr). Combined saving **12.0%** (182 t/yr, €118k/yr). The effective SFOC of the main engine for the SG load is $225\,\mathrm{g/kWh}$ — close to a typical auxiliary genset at $215\,\mathrm{g/kWh}$ — making the SG vs genset trade-off marginal at $10\,\mathrm{kn}$. A separate genset saves approximately $11\,\mathrm{t/yr}$ (€7k/yr).

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3. The factory combinator becomes **infeasible** (engine over-power) on 7% of voyages, while the optimiser adapts pitch/RPM to remain within the power envelope.
4. The **modular architecture** (4 packages, 24 modules, 5 181 lines) provides clean separation of physical models, simulation logic, reporting, and visualisation, enabling independent testing and extension of each component.

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